

Assignment 11 Recommender System

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Introduction

Scenario Design is a user-centric framework that shifts the focus from building features to fulfilling fundamental user goals. Instead of asking “What can we build?”, it asks “What does our user need to achieve?” This analysis applies this mindset to reverse-engineer the recommender systems of **Netflix** and **Duolingo**, exploring how they serve their users’ deepest needs for effortless entertainment and meaningful learning.

Scenario Design Analysis: Netflix

Identified Recommender System: Netflix: <https://www.netflix.com>

Answering the Three Scenario Design Questions 1. Who are your target users?

Netflix’s target users for its recommender system are **subscribers at various stages of engagement and viewing habits**. This includes:

- The **“I don’t know what to watch” user** who feels overwhelmed by choice.
- The **binge-watcher** who has just finished a series and needs the next one.
- The **mood-based viewer** looking for something specific like a “lighthearted comedy” or a “gripping thriller.”
- The **multi-profile household** where a single account must serve the diverse tastes of different family members.

2. What are their key goals?

The fundamental user goals are not just to “play a video,” but to:

- **Minimize Decision Fatigue & Time Spent Browsing:** Users want to find something enjoyable quickly, without the frustration of scrolling through endless rows.
- **Discover Content They Will Love:** They want to be surprised and delighted by hidden gems and new releases tailored to their taste.
- **Maintain a Continuous, Engaging Viewing Experience:** The goal is to seamlessly move from one satisfying piece of content to the next, maintaining their engagement and subscription.
- **Have a Personalized Experience:** In a shared account, each user wants to feel that the homepage is “theirs,” reflecting their individual preferences.

3. How can you help them accomplish those goals?

Instead of just building a “similar movies” algorithm, Netflix’s recommender system is designed to serve these user goals through a multi-faceted approach:

- **To Minimize Decision Fatigue:** The system provides highly curated rows like “**Top Picks for [User]**” and the prominent “**Match**” **score percentage** on each title. This gives users a quick, confidence-inspiring answer to “what should I watch?”
- **To Facilitate Discovery:** The interface is built around discovery engines like “**Because you watched [X]**” and “**Trending Now.**” This provides a logical reason for recommendations and helps users explore related genres, directors, or actors they’ve enjoyed before.
- **To Maintain Engagement:** The **auto-play trailer feature** and the “**Post-Play**” countdown that automatically starts the next episode (or suggests a new show) are designed to reduce friction and make continuous viewing the path of least resistance.
- **To Enable Personalization:** The core of the entire system is the **individual user profile**. By separating viewing histories and ratings, the recommender engine can build a unique model for each person on the account, ensuring that a child’s cartoon binge doesn’t affect a parent’s recommendations for documentaries.

The Mindset Shift: From Functionality to Goals

- **Before Scenario Design (Functionality-First):** “We should offer a filtering system by genre, a search bar, and a ‘most popular’ list.”
- **After Scenario Design (Goal-First):** “How can we create a personalized homepage that makes a user feel understood, so they can find a movie they’ll enjoy in under 60 seconds?”

By focusing on the user goals of quick, satisfying discovery and continuous engagement, Netflix has built a recommender system that is central to its value proposition and massive customer retention. The various features (rows, match scores, auto-play) are not the starting point; they are the solutions designed to fulfill these deeper human needs.

Scenario Design Analysis: Duolingo

Identified Recommender System: Duolingo (specifically, its systems for recommending practice lessons, new skills, and review activities) <https://www.duolingo.com>

Answering the Three Scenario Design Questions 1. Who are your target users?

Duolingo’s target users are **language learners with varying motivations and consistency levels**. This includes:

- The “**Casual Dabbler**” who is learning for fun or travel and uses the app intermittently.
- The “**Serious Student**” who has a clear goal (e.g., career advancement, passing an exam) and seeks structured progress.
- The “**Lapsed Learner**” who has skipped several days and is at high risk of quitting entirely.
- The “**Competitive Learner**” who is motivated by gamification elements like streaks and leaderboards.

2. What are their key goals?

The fundamental user goals are not just to “complete a lesson,” but to:

- **Make Tangible, Feelable Progress:** Users want to feel they are actually getting better at understanding and speaking the language.
- **Build a Consistent Habit:** The primary challenge in language learning is consistency. Users want to integrate practice seamlessly into their daily routine.
- **Overcome the “Forgetting Curve”:** Users need to review old material at the optimal time to move knowledge from short-term to long-term memory.

- **Stay Motivated and Avoid Burnout:** The learning path should feel engaging, not overwhelming or repetitive.

3. How can you help them accomplish those goals?

Instead of just presenting a linear path of lessons, Duolingo’s recommender systems are designed to serve these user goals through adaptive, personalized interventions:

- **To Make Tangible Progress:** The “**Personalized Practice**” section doesn’t just randomly review old words. It uses a predictive model to identify the specific words and grammar rules a user is about to forget and creates a targeted review session. This makes practice efficient and shows the user they are actively strengthening their weak points.
- **To Build a Consistent Habit:** The entire app is a recommender system for daily use. The **Daily Goal notifications**, the emphasis on **maintaining a streak**, and the “**perfect**” and “**hard**” **practice buttons** that appear after a lesson are all designed to recommend the next micro-action to keep the habit loop going.
- **To Overcome the Forgetting Curve:** This is the core of Duolingo’s recommendation engine. The backend employs a **Spaced Repetition System (SRS)** algorithm. This algorithm predicts when a user is most likely to forget a specific vocabulary item and proactively surfaces it for review just before that point, making practice incredibly efficient and effective.
- **To Maintain Motivation:** The system recommends actions based on user psychology. For a “Competitive Learner,” it might highlight the **Leaderboard** or a “Friend Quest.” For a “Lapsed Learner,” it sends a compassionate “We miss you!” notification and recommends a simple, 5-minute “review” lesson to get them back on track without pressure. It avoids recommending overly difficult new skills right after a break, preventing frustration and burnout.

The Mindset Shift: From Functionality to Goals

- **Before Scenario Design (Functionality-First):** “We should offer a list of vocabulary lessons, a grammar guide, and a multiple-choice quiz.”
- **After Scenario Design (Goal-First):** “How can we create a personalized learning coach that identifies what a user is about to forget, makes review feel urgent and rewarding, and integrates a daily learning habit into their life?”

By focusing on the user goals of measurable progress, habit formation, and long-term retention, Duolingo’s recommender systems act as an adaptive tutor. The features (practice sessions, notifications, review queues) are not arbitrary; they are tactical solutions designed to solve the fundamental human challenges of learning a new skill over time.

Conclusion: From Utility to Empathy in Recommender Systems

The Scenario Design analyses of Netflix and Duolingo reveal a crucial evolution in the philosophy behind recommender systems. Both platforms have moved beyond treating their systems as mere functional utilities for sorting content. Instead, they have engineered them as **empathic, goal-oriented partners**.

Netflix understands that its users’ primary goal is not to *find* a video, but to *relax and be entertained* without the mental burden of choice. Its recommender system is therefore designed as a solution to **decision fatigue**, prioritizing seamless discovery and continuous engagement to make leisure time truly effortless.

Similarly, Duolingo recognizes that its users’ goal is not merely to complete lessons, but to **achieve real, lasting language proficiency**. Its system acts as a personal tutor, combating the natural tendency to forget and the struggle to build a habit. It recommends reviews and actions specifically tailored to sustain motivation and ensure tangible progress.

In both cases, the success of the recommender is not just in its algorithmic accuracy, but in its deep alignment with fundamental human needs: the desire for effortless enjoyment and the aspiration for personal growth.

By asking, “**What user goals should we serve?**” instead of “**What functionality should we offer?**”, Netflix and Duolingo have created indispensable services that are deeply woven into the daily lives of their users.